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Rumor Has It: Sensationalism in Financial Media

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1 Big picture

- **Question.** Does the business press trade off accuracy for readership appeal, and do investors understand this trade-off when reacting to financial news?
- **Setting.** The paper studies merger rumors in newspapers from 2000–2011. Merger rumors are useful because the researcher can observe an ex post outcome: whether a formal takeover bid is announced after the rumor.
- **Core idea.** Newspapers compete for readers. That gives them an incentive to publish sensational stories about large, familiar, and attention-grabbing firms. These stories may attract readers even when they are less likely to be accurate.
- **Main empirical object.** The authors hand-collect the first newspaper article that reports a rumor, called the **scoop article**. They then ask whether the rumor comes true within one year.
- **Main result on media accuracy.** Rumors are less accurate when they involve more newsworthy firms: large public targets, valuable brands, and firms with high advertising. Rumors are more accurate when written by older journalists, journalists with journalism training, and journalists who specialize in the target’s industry.
- **Main result on article text.** The words and details in the article predict accuracy. Articles using more weak modal language, such as words like “maybe” and “appears,” are less likely to come true. Articles that mention a takeover price, potential bidders, or concrete negotiation stages are more likely to come true.
- **Main result on asset prices.** Stock prices react strongly to rumors, but investors do not fully use the information that predicts rumor accuracy. Targets earn a large positive return on the rumor date, but inaccurate rumors subsequently reverse.
- **Limited attention channel.** Public information about journalists, target newsworthiness, and article language predicts future returns, but is not fully incorporated into the immediate price response. This is consistent with investors paying attention to the headline news but not fully processing all credibility signals.
- **Limits to arbitrage.** Mispricing is larger and reverses more slowly among stocks that are harder to arbitrage, such as illiquid stocks and stocks with high idiosyncratic volatility.
- **Real effects.** Merger rumors affect the timing and pricing discussion in actual deals. However, acquirers appear to discount the rumor-induced run-up: rumored deals have higher pre-announcement run-ups but lower announcement markups, leaving total takeover premiums roughly unchanged.
- **Takeaway.** The media is not just a neutral transmitter of information. Its incentive to attract attention shapes which stories are published, how credible they are, and how prices respond.

2 Data and measurement (key objects)

2.1 Sample construction

The sample is built from newspaper merger rumors. The authors search Factiva for articles in major U.S. news and business publications from January 1, 2000 through December 31, 2011. Search terms combine merger-related words such as *acquire*, *merger*, *takeover*, *buyout*, or *bid* with rumor-related words such as *rumor*, *speculation*, *said to be*, or *talks*.

The construction has two steps:

1. identify articles that report merger rumors and the target firms named in them;
2. trace the rumor backward to find the earliest article reporting the rumor, called the **scoop article**.

The final dataset includes:

Object	Count or value
All rumor articles	2,142
Unique merger rumors / scoop articles	501
Unique target firms	354
Rumors followed by public bid within one year	167
Baseline accuracy rate	33.3%

Interpretation. The sample is small relative to all SDC mergers, but it covers economically large firms. The aggregate market equity of rumor targets is a large fraction of the aggregate value of all acquisitions during the period.

2.2 Definition of accuracy

The central dependent variable is:

$$\text{Accuracy}_i = \mathbf{1}\{\text{target } i \text{ receives a public takeover bid within one year of the rumor}\}. \quad (1)$$

This is not a literal definition of truth. A rumor could be literally true because someone privately discussed a takeover, yet no official bid ever appears. The paper uses a reader-centered definition: a rumor is accurate if it predicts an economically meaningful future event.

Alternative accuracy definitions are also considered in robustness checks:

- bid within one month, three months, six months, or nine months;
- bid within one year and correct identification of the bidder;
- shorter time lags between rumor and announcement.

Important measurement point. The baseline definition is generous. It does not require the article to name the true bidder and gives the rumor one full year to be validated.

2.3 Target newsworthiness variables

The paper measures whether a rumor target has broad readership appeal. The variables are motivated by journalism concepts of **breadth**, **prominence**, and **proximity**.

Key variables include:

- **Public target**: whether the target is publicly traded.
- **Log target book assets**: firm size.
- **Valuable brand**: whether the firm appears in Interbrand or BrandZ top-brand lists.
- **Advertising/assets**: public visibility and consumer recognition.
- **Industry sales to households**: how much the target's industry sells directly to households.
- **Tobin's q and R&D/assets**: growth and innovation.
- **Distance**: distance between the newspaper and target headquarters.
- **Foreign target**: whether the target is headquartered outside the United States.

Summary statistics show that rumor targets are unusually newsworthy: about 88% are public, 16% have valuable brands, and average book assets are large.

2.4 Journalist characteristics

The authors hand-collect biographical data on journalists who write scoop articles. The dataset includes 382 journalists.

Important journalist variables are:

- average journalist age;
- undergraduate major, including journalism, business and economics, English, political science, history, and other;
- college quality, measured using verbal SAT percentile;
- whether at least one author specializes in the target's industry;
- whether at least one author is based in New York;
- whether the journalist has won or been nominated for major journalism awards;
- journalist gender.

Measurement interpretation. These variables proxy for a journalist's ability to filter information. Experience, professional training, industry specialization, and Wall Street proximity may help a journalist distinguish credible leads from idle speculation.

2.5 Article text and rumor details

The paper codes features of each scoop article.

Language ambiguity. The main text variable is the percentage of weak modal words. The dictionary comes from Loughran and McDonald and includes words such as *apparently*, *maybe*, *perhaps*, and *suggests*. More weak modal language indicates lower author confidence.

Rumor details. The authors also code:

- whether the source is anonymous;
- whether the target declined to comment, confirmed, denied, could not be reached, or was not asked;
- the stage of the merger talks, from speculation to evaluating bids;
- whether the rumor appears in the headline;
- number of potential bidders mentioned;
- whether a takeover price is mentioned;
- number of rumor articles on the scoop date.

Interpretation. The article can contain both sensational signals and credibility signals. A story may attract attention, but its exact wording and specificity reveal information about whether the journalist has credible evidence.

2.6 Newspaper characteristics

The paper also collects newspaper-level variables:

- newspaper age;
- circulation;
- ownership type, especially whether the parent company is family-run;
- newspaper fixed effects.

Newspaper identity is easier for readers to observe than individual journalist biographies. The paper later shows that newspaper characteristics are less predictive of accuracy than journalist and article characteristics.

2.7 Stock returns and trading variables

For public targets, abnormal returns are calculated using CRSP:

$$AR_{i,t} = r_{i,t} - r_{m,t}^{VW}, \quad (2)$$

where $r_{m,t}^{VW}$ is the CRSP value-weighted market return.

Cumulative abnormal returns are:

$$CAR_i[a, b] = \sum_{t=a}^b AR_{i,t}. \quad (3)$$

Important event windows are:

- $[-20, -1]$, $[-10, -1]$, and $[-5, -1]$: pre-rumor run-up;
- day 0: rumor publication date;
- $[+1, +5]$, $[+1, +10]$, and $[+1, +20]$: post-rumor reversal;
- $[-20, +20]$: full event window around the rumor.

The paper also studies:

- short-selling utilization using Markit equity lending data;
- institutional trading using ANcerno data;
- liquidity and idiosyncratic volatility as limits-to-arbitrage variables.

3 Models and empirical specifications (all equations + interpretation)

3.1 Event-study return measures

The basic event-study objects are abnormal returns and cumulative abnormal returns:

$$\text{AR}_{i,t} = r_{i,t} - r_{m,t}^{VW}, \quad (4)$$

$$\text{CAR}_i[a, b] = \sum_{t=a}^b \text{AR}_{i,t}. \quad (5)$$

Interpretation. If investors immediately price the probability that a rumor is true, accurate rumors should have larger day-0 returns than inaccurate rumors. If investors overreact to rumors on average, inaccurate rumors should reverse after publication.

3.2 Baseline accuracy logit

The main accuracy specification is a logit regression:

$$\Pr(\text{Accuracy}_i = 1) = \Lambda(\alpha + X_i' \beta + \gamma \text{CAR}_i[-5, -1] + \delta_{\text{industry}} + \tau_{\text{year}}), \quad (6)$$

where X_i is one of the paper's sets of predictive variables:

- target newsworthiness characteristics;
- journalist characteristics;
- article characteristics;
- newspaper characteristics or fixed effects.

The term $\text{CAR}_i[-5, -1]$ controls for stale information or information leakage before the article

appears.

Interpretation. The coefficients β answer the first main question: which public features of the rumor predict whether it comes true?

3.3 Adding the market reaction

The second specification adds the target's day-0 abnormal return:

$$\Pr(\text{Accuracy}_i = 1) = \Lambda(\alpha + X_i' \beta + \eta \text{AR}_{i,0} + \gamma \text{CAR}_i[-5, -1] + \delta_{\text{industry}} + \tau_{\text{year}}). \quad (7)$$

Interpretation. If day-0 stock returns fully reflect all public signals about accuracy, then variables in X_i should no longer predict accuracy after controlling for $\text{AR}_{i,0}$. If X_i remains significant, the market did not fully process those credibility signals.

3.4 Estimated deal likelihood

The paper refines the day-0 return control using the idea:

$$r_{0,i} = p_i \times r_{a,i}, \quad (8)$$

where p_i is the probability that the rumor comes true and $r_{a,i}$ is the target's expected return if a public merger announcement occurs.

Rearranging gives:

$$p_i = \frac{r_{0,i}}{r_{a,i}}. \quad (9)$$

The authors estimate $r_{a,i}$ from a sample of official merger announcements and construct:

$$\hat{p}_i = \frac{r_{0,i}}{\hat{r}_{a,i}}. \quad (10)$$

Then they estimate:

$$\Pr(\text{Accuracy}_i = 1) = \Lambda(\alpha + X_i' \beta + \eta \hat{p}_i + \gamma \text{CAR}_i[-5, -1] + \delta_{\text{industry}} + \tau_{\text{year}}). \quad (11)$$

Interpretation. The day-0 return reflects both the probability of a bid and the expected size of the bid premium. The estimated deal likelihood tries to isolate the probability component.

3.5 Day-0 return regression

The paper also asks which signals investors appear to react to on the publication date. The specification is:

$$\text{AR}_{i,0} = \alpha + X_i' \beta + \theta \hat{r}_{a,i} + \gamma \text{CAR}_i[-5, -1] + \varepsilon_i. \quad (12)$$

Interpretation. The logit regressions show what predicts true accuracy. The day-0 return regression shows what the market appears to price. Comparing the two reveals which signals investors use and which they neglect.

3.6 Long-short portfolio tests

The paper forms portfolios using predicted accuracy. Let $\widehat{\text{Accuracy}}_i$ be the fitted value from one of the accuracy models. The strategy is:

$$\text{Long}_t = \{i : \widehat{\text{Accuracy}}_i > \text{median}(\widehat{\text{Accuracy}})\}, \quad (13)$$

$$\text{Short}_t = \{i : \widehat{\text{Accuracy}}_i \leq \text{median}(\widehat{\text{Accuracy}})\}. \quad (14)$$

The long-short return is:

$$R_t^{LS} = R_t^{\text{long}} - R_t^{\text{short}}. \quad (15)$$

Firms enter the portfolio on the day after the rumor. This means the day-0 return is not included.

Interpretation. These portfolios are not meant to be directly implementable trading strategies because they use full-sample information and do not fully account for transaction costs. They quantify how much return predictability remains when investors fail to process accuracy signals.

3.7 Limits-to-arbitrage tests

The paper splits the sample by measures of arbitrage difficulty:

- Amihud illiquidity;
- idiosyncratic volatility;
- bid-ask spreads.

The prediction is:

$$R_{\text{hard to arbitrage}}^{LS} > R_{\text{easy to arbitrage}}^{LS}. \quad (16)$$

Interpretation. If mispricing is real but arbitrage is costly, return predictability should be strongest for stocks where trading against the overreaction is difficult.

3.8 Short-selling and institutional trading

Short utilization is measured as:

$$\text{ShortUtilization}_{i,t} = \frac{\text{SharesOnLoan}_{i,t}}{\text{SharesLendable}_{i,t}}. \quad (17)$$

Abnormal short utilization subtracts a pre-rumor benchmark:

$$\text{AbShortUtil}_{i,t} = \text{ShortUtilization}_{i,t} - \overline{\text{ShortUtilization}}_{i,\text{pre}}. \quad (18)$$

Institutional buy-sell imbalance is:

$$\text{BSI}_{i,t} = \frac{\text{InstitutionalBuys}_{i,t} - \text{InstitutionalSells}_{i,t}}{\text{InstitutionalBuys}_{i,t} + \text{InstitutionalSells}_{i,t}}. \quad (19)$$

Interpretation. If sophisticated investors process rumor credibility better than retail investors, short sellers should reduce short positions in accurate rumors, and institutions should be less likely to be the net buyers driving rumor-date price increases.

3.9 Real-effects regressions

For official merger bids, the paper estimates regressions of target returns around the announcement:

$$\text{CAR}_{i,w} = \alpha + \rho \text{Rumor}_i + Z_i' \beta + \delta_{\text{industry}} + \tau_{\text{year}} + \varepsilon_i, \quad (20)$$

where w is one of three windows:

- $[-42, -1]$: run-up before public merger announcement;
- $[0, +5]$: announcement markup;
- $[-42, +5]$: total premium window.

Interpretation. If rumors simply increase the price paid by acquirers, the total premium should be higher for rumored deals. If acquirers understand and discount the rumor-induced run-up, then run-up returns should be higher, announcement returns should be lower, and total premiums should be similar.

4 Identification and credibility

4.1 Why merger rumors are a useful setting

The empirical difficulty in studying media accuracy is that the truth is usually unobservable. Merger rumors solve part of this problem because the researcher can observe whether an official takeover bid follows the rumor.

The setting is also useful because rumors are financially important. Target stock prices react strongly to merger news, so mistakes in perceived credibility can generate measurable asset-pricing effects.

4.2 Selection into media coverage

The paper's first selection issue is that the researcher observes published rumors, not unpublished rumors. The authors address this by comparing rumor targets to actual merger targets.

The comparison shows that published rumors are concentrated among more newsworthy firms: large public firms, familiar brands, domestic firms, and firms with advertising visibility. This supports the media-incentive interpretation: newspapers choose stories that readers want to read.

Limitation. Because unpublished rumors are unobserved, the paper cannot directly compare published and unpublished rumors. It can show that published rumors differ from realized mergers and that newsworthiness predicts lower accuracy within the published-rumor sample.

4.3 Measuring truth with future bids

The dependent variable is not literal truth. A rumor can be based on real conversations that never become a public bid. The paper's definition is economically useful because it captures what readers and investors care about: whether a takeover materializes.

The paper addresses this concern in two ways:

- it uses alternative windows for accuracy and shows the main results are robust;
- it tests whether more newsworthy firms are more likely to have failed negotiations by studying withdrawals of public bids, and finds no evidence that newsworthiness predicts withdrawals.

4.4 Separating investor attention from information leakage

A rumor may circulate before it appears in the newspaper. If so, the day-0 stock reaction may understate the information content of the article. The paper controls for this using pre-rumor abnormal returns, especially $CAR[-5, -1]$.

This control is important because it helps distinguish two explanations:

- the market ignores credibility signals;
- the market already learned the information before the article appeared.

The persistence of the results after controlling for pre-publication returns supports the limited-attention interpretation.

4.5 Market reaction as a benchmark for investor beliefs

The paper's most important identification logic is the comparison between accuracy models and day-0 return models.

- If a variable predicts accuracy and also predicts day-0 returns, investors appear to price that signal.
- If a variable predicts accuracy but does not predict day-0 returns, investors appear to overlook that signal.
- If a variable predicts day-0 returns but not accuracy, investors react to a signal that is not actually informative.

This is why Tables 6–8 are central: they put the same variables into both the truth model and the price-reaction model.

4.6 Why journalist characteristics matter

Journalist characteristics are useful because they are publicly observable in principle but not necessarily salient. Readers may see the byline, but they may not know the journalist's age, education, industry specialization, or historical accuracy.

The paper finds that these characteristics predict accuracy even after controlling for the day-0 return. This suggests that the market does not fully process author-level credibility.

4.7 Limits to arbitrage and return predictability

Even if some investors recognize that a rumor is likely inaccurate, mispricing may persist if arbitrage is costly. Shorting a stock after a merger rumor can be risky because a true takeover bid creates a large loss for short sellers.

The paper's limits-to-arbitrage tests support this logic. Return predictability is stronger and reversals are slower for illiquid and high-idiosyncratic-volatility stocks.

4.8 Alternative explanations

The paper addresses several alternative explanations:

- **Actual merger activity.** Rumor frequency is not tightly tied to aggregate merger activity.
- **Extreme prior returns.** Newsworthy firms are not simply selected because they had extreme pre-rumor stock movements.
- **Bidder newsworthiness.** Bidders are also newsworthy, but target newsworthiness is more strongly related to coverage and price reactions.
- **Failed negotiations.** More newsworthy targets are not more likely to have public bids withdrawn.
- **Newspaper reputation.** Newspaper identity matters less for accuracy than journalist traits and article text.

4.9 Main limitations

The paper is persuasive but not without limits:

- It cannot observe all private merger rumors that never reach the press.
- The accuracy definition may misclassify true private talks that fail before an official bid.
- The long-short portfolios are in-sample and not directly implementable trading strategies.
- The setting is merger rumors, so external validity to other financial news requires caution.
- Some journalist variables may proxy for unobserved newsroom assignment or source quality.

Bottom line. The identification is strongest for the claim that observable features of published rumors predict ex post bid announcements and are not fully reflected in prices. It is more suggestive, but still compelling, on the deeper mechanism of media incentives.

5 Tables and figures

5.1 Table 1: Rumor articles by calendar period

Read. The sample contains 2,142 rumor articles and 501 scoop articles from 2000–2011. Of the 501 rumors, 167 come true within one year, giving a baseline accuracy rate of 33.3%.

Selected numbers:

Period / total	All articles	Scoop articles	Accuracy rate
Full sample	2,142	501	33.3%
2010	393	75	34.7%
2007	279	70	35.7%
2004	58	14	28.6%

Economic message. Merger rumors are frequent and financially important, but only about one-third are followed by a bid. Rumors are not just previews of actual merger waves; their frequency is only weakly related to formal merger activity.

5.2 Table 2: Summary statistics of predictive variables

Read. Rumor targets and articles have strong attention-grabbing features:

Variable	Mean	Interpretation
Public target	0.884	Most rumor targets are public firms
Valuable brand	0.156	Brand-name firms are overrepresented
Industry sales to households	0.384	Targets often sell to consumers
Expert in target industry	0.554	Many authors have industry expertise
New York–based journalist	0.495	Nearly half involve New York journalists
Weak modal words	0.745%	Rumor articles use speculative language
Anonymous source	0.920	Most stories rely on anonymous sources
Price mentioned	0.386	Some stories give concrete deal-price details

Economic message. The dataset is designed to separate several credibility channels: target appeal, journalist skill, article specificity, and newspaper reputation.

5.3 Table 3: Accuracy rates of journalists and media sources

Read. Accuracy varies widely across journalists and outlets. Dennis Berman has 24 scoop articles and a 62.5% accuracy rate. Andrew Ross Sorkin has 19 scoop articles and a 42.1% accuracy rate. At the outlet level, the Wall Street Journal has 158 scoops and a 38.6% accuracy rate; Dow Jones News Service has 67 scoops and a 38.8% accuracy rate; the full sample accuracy rate is 33.3%.

Economic message. The identity of the person reporting the rumor matters. Some journalists are much more accurate than average, and this variation becomes central to the paper’s limited-attention evidence.

Table 3
Accuracy rates of journalists and media sources

Journalist/Media source	All articles	Percentage of all articles	Scoop articles	Percentage of scoops	Accuracy rate
Panel A: Journalists					
Dennis K. Berman (WSJ)	71	3.2	24	4.0	62.5
Andrew Ross Sorkin (NYT)	67	3.0	19	3.2	42.1
Nikhil Deogun (WSJ)	27	1.2	13	2.2	53.8
Robert Frank (WSJ)	20	0.9	13	2.2	23.1
Robin Sidel (WSJ)	23	1.0	9	1.5	55.6
Anupreet Das (WSJ)	21	0.9	7	1.2	42.9
Michael J. de la Merced (NYT)	32	1.4	6	1.0	16.7
Jeffrey McCracken (Bloomberg)	19	0.9	6	1.0	50.0
Anita Raghavan (NYT)	19	0.9	6	1.0	16.7
Suzanne Kapner (NYT)	18	0.8	6	1.0	33.3
Sarah Ellison (DJNS)	16	0.7	6	1.0	33.3
Erica Copulsky (NY Post)	14	0.6	6	1.0	33.3
786 others	1,867	84.3	482	79.9	36.3
Total	2,214	100.0	603	100.0	37.6
Panel B: Media sources					
Wall Street Journal	448	20.9	158	31.5	38.6
Dow Jones News Service	625	29.2	67	13.4	38.8
New York Times	219	10.2	38	7.6	28.9
Reuters News	73	3.4	26	5.2	19.2
New York Post	95	4.4	24	4.8	37.5
Barron's	38	1.8	15	3.0	26.7
NYT Blogs	59	2.8	12	2.4	16.7
Bloomberg	16	0.7	10	2.0	80.0
Boston Globe	38	1.8	8	1.6	25.0
Financial Times	15	0.7	8	1.6	62.5
Los Angeles Times	7	0.3	6	1.2	16.7
94 others	509	23.8	129	25.7	25.2
Total	2,142	100.0	501	100.0	33.3

This table presents publishing activity for journalists and media sources in the sample from 2000 to 2011. The scoop article is the newspaper article that first reports the merger rumor. Accuracy rate is the fraction of scoop articles in which a formal takeover bid is made for the target firm within one year. The total number of articles and scoops for journalists exceeds the total number for media sources because some articles include multiple authors. Abbreviations in parentheses indicate the most recent newspaper that employed a journalist. *WSJ* indicates *Wall Street Journal*, *NYT* indicates *New York Times*, *DJNS* indicates *Dow Jones News Service*, and *NY Post* indicates *New York Post*.

5.4 Table 4: Target characteristics in rumors versus actual mergers

Read. Rumor targets are much more newsworthy than actual merger targets:

Characteristic	Rumor targets	All merger targets
Public target	88.42%	38.10%
Valuable brand	15.57%	0.48%
Advertising/assets	0.83%	0.14%
Industry sales to households	38.41%	30.53%
Foreign target	25.15%	66.08%

Economic message. Published rumors are not a random sample of potential takeovers. They disproportionately feature firms that readers recognize and care about. This supports the idea that media coverage reflects demand for attention-grabbing stories.

5.5 Table 5 and Figure 1: Target abnormal event returns and reversals

Read. On the rumor publication date, targets earn large abnormal returns. Accurate rumors generate larger day-0 returns than inaccurate rumors, but inaccurate rumors reverse afterward.

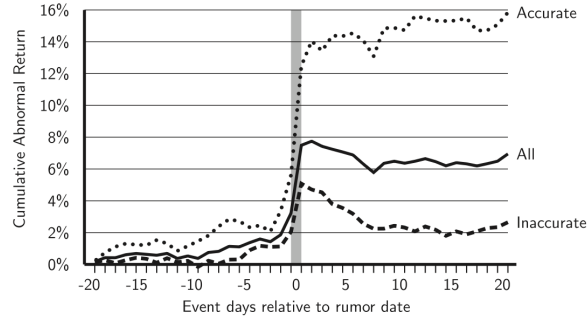


Figure 1
Abnormal returns of accurate and inaccurate merger rumors
This figure presents the time series of cumulative abnormal returns of merger rumor targets for three time series relative to the date of the first publication of a merger rumor. There are 414 rumors with stock price data over 2000–2011. Cumulative abnormal returns are the cumulative sum of daily abnormal returns, calculated as the firm's daily return minus the value-weighted CRSP index. The dotted line represents the subsample of merger rumor targets where a public merger announcement was made within one year of the rumor date. The dashed line represents the subsample of merger rumor targets where no public merger announcement was made in the following year.

Table 6
Rumor accuracy and stock returns: Target newsworthiness

Dependent variable:	Rumor comes true			Return day 0
	(1)	(2)	(3)	(4)
Day 0 return		4.458*** (< 0.001)		
Estimated deal likelihood			0.282*** (< 0.001)	
Estimated announcement return				0.083 (0.393)
Log(Target book assets)	−0.287*** (0.004)	−0.264*** (0.005)	−0.279*** (0.007)	
Valuable brand	−0.500* (0.078)	−0.382* (0.082)	−0.461* (0.077)	−0.042*** (0.006)
Advertising/Assets (%)	−0.068** (0.035)	−0.073*** (0.007)	−0.069** (0.021)	0.001 (0.796)
Industry sales to households	0.323 (0.315)	0.377 (0.351)	0.432 (0.196)	0.002 (0.901)
Tobin's q	−0.045 (0.206)	−0.022 (0.574)	−0.030 (0.420)	−0.005** (0.020)
R&D/Assets (%)	0.007 (0.847)	−0.005 (0.860)	0.004 (0.910)	0.003** (0.022)
Log(1+Distance)	0.006 (0.729)	−0.013 (0.475)	−0.010 (0.619)	0.003** (0.020)
Returns _(−5,−1)	0.012 (0.990)	1.500 (0.271)	0.553 (0.548)	−0.279** (0.034)
Industry fixed effects	yes	yes	yes	no
Year fixed effects	yes	yes	yes	no
Observations	399	399	399	399
Pseudo/Adjusted R^2	0.103	0.138	0.126	0.112

This table examines the relationship between target newsworthiness and (1) the likelihood that a rumor comes true and (2) the target stock returns on the day the rumor is first published. Columns 1–3 present fixed effect logit regression coefficients in which the dependent variable is a dummy variable equal to one if a merger rumor came true within one year of the first rumor date. Column 4 presents OLS regression coefficients where the dependent variable is the rumor target's abnormal stock return on the date the first rumor article is published. Abnormal returns are calculated as the target returns minus the CRSP value-weighted index return. Variables are defined in the appendix. Industry fixed effects include Fama-French 17 industry codes. Standard errors are clustered at the industry level, and p -values are reported in parentheses. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

Event window	All rumors	Accurate rumors	Inaccurate rumors
Day 0	4.271%	6.865%	3.029%
Days $[-20, -1]$	3.093%	5.267%	2.045%
Days $[+1, +10]$	−1.395%	1.422%	−2.743%
Days $[-20, +20]$	6.550%	14.736%	2.633%

Economic message. Investors react positively to merger rumors, but they overreact to the average rumor. Figure 1 visualizes this: accurate rumors continue to perform well, while inaccurate rumors partially reverse.

5.6 Table 6: Rumor accuracy and target newsworthiness

Read. Newsworthiness predicts lower accuracy. In the baseline logit, log target assets, valuable brand, and advertising/assets have negative coefficients. These effects remain after controlling for day-0 returns or estimated deal likelihood.

Key interpretation from the paper:

- the odds that a rumor comes true are 1.65 times larger for firms without a valuable brand than for firms with a valuable brand;
- a one-standard-deviation increase in log assets reduces the odds of accuracy by about 43%.

Economic message. The same firm features that attract readers are associated with lower rumor accuracy. Investors do not fully discount this reduced credibility.

5.7 Table 7: Rumor accuracy and journalist characteristics

Read. Journalist traits strongly predict accuracy. Older journalists, journalists with journalism degrees, and journalists with target-industry expertise are more accurate. New York-based journalists are also more accurate in the baseline specification.

Selected baseline logit coefficients:

Variable	Coefficient in accuracy logit
Log journalist age	1.378
Journalism major	1.177
Expert in target industry	0.592
New York-based	0.440

Economic message. Accuracy depends on human capital and source networks. The market does not fully price the accuracy content of journalist age, journalism education, or industry expertise.

5.8 Table 8: Rumor accuracy and article characteristics

Read. Textual features predict whether a rumor comes true:

- weak modal words have a negative coefficient of about -0.834 in the baseline accuracy logit;
- rumors saying the firms are “in talks” have a positive coefficient of about 1.319;
- more named bidders and a mentioned takeover price are positive predictors of accuracy;
- these patterns remain after controlling for day-0 returns.

Economic message. The article itself contains credibility signals. Concrete details predict truth; ambiguous language predicts inaccuracy. Investors appear to react to the headline, but do not fully parse the article’s credibility content.

Table 8
Rumor accuracy and stock returns: Article characteristics

Dependent variable:	Rumor comes true			Return day 0
	(1)	(2)	(3)	(4)
Day 0 return		4.582*** (< 0.001)		
Estimated deal likelihood			0.273*** (0.001)	
Estimated announcement return				0.099 (0.118)
Weak modal words (%)	−0.834*** (0.001)	−0.877*** (0.003)	−0.857*** (0.001)	0.005 (0.444)
Anonymous source	0.367 (0.603)	0.306 (0.676)	0.366 (0.602)	0.019*** (0.004)
<i>Target response</i>				
Has conversations	−0.193 (0.718)	−0.040 (0.942)	−0.041 (0.940)	−0.007 (0.529)
Confirmed rumor	1.234*** (0.010)	0.975* (0.081)	1.119** (0.026)	0.076** (0.016)
Denied rumor	−0.999 (0.271)	−1.068 (0.246)	−0.984 (0.250)	0.007 (0.655)
Couldn't be reached	0.446 (0.209)	0.259 (0.458)	0.403 (0.250)	0.048*** (< 0.001)
Wasn't asked	−0.101 (0.828)	−0.158 (0.731)	−0.055 (0.903)	0.014 (0.165)
<i>Merger stage</i>				
Preliminary talks	0.728** (0.019)	0.599* (0.061)	0.620* (0.060)	0.014 (0.312)
In talks	1.319*** (0.001)	1.340*** (< 0.001)	1.319*** (0.001)	−0.003 (0.785)
Made offer	0.611 (0.505)	0.604 (0.487)	0.684 (0.509)	−0.003 (0.820)
Preparing bid	0.648 (0.488)	0.582 (0.520)	0.536 (0.571)	0.017 (0.613)
For sale	0.361 (0.461)	0.459 (0.385)	0.590 (0.201)	−0.015 (0.406)
Evaluating bids	0.965 (0.455)	1.057 (0.407)	0.959 (0.449)	−0.022 (0.299)
Articles on scoop date (#)	0.164** (0.043)	0.057 (0.437)	0.107 (0.176)	0.022** (0.031)
Rumor in headline	0.002 (0.998)	0.019 (0.978)	−0.051 (0.940)	0.000 (0.992)
Number of bidders mentioned	0.098** (0.011)	0.101** (0.036)	0.101*** (0.010)	−0.001 (0.763)
Price mentioned	0.667*** (0.002)	0.674*** (0.005)	0.634*** (0.005)	0.008 (0.487)
Returns _(−5,−1)	0.013 (0.982)	1.229* (0.068)	0.442 (0.496)	−0.266* (0.056)
Log(Target book assets)	−0.273*** (0.004)	−0.242** (0.020)	−0.260** (0.015)	
Industry fixed effects	yes	yes	yes	no
Year fixed effects	yes	yes	yes	no
Observations	372	372	372	372
Pseudo/Adjusted R^2	0.224	0.244	0.237	0.202

This table examines the relationship between article characteristics and (1) the likelihood that a rumor comes true and (2) the target stock returns on the day the rumor is first published. Columns 1–3 present fixed effect logit regression coefficients in which the dependent variable is a dummy variable equal to one if a merger rumor came true within one year of the first rumor date. Column 4 presents OLS regression coefficients where the dependent variable is the rumor target's abnormal stock return on the date the first rumor article is published. Abnormal returns are calculated as the target returns minus the CRSP value-weighted index return. Variables are defined in the appendix. Industry fixed effects include Fama-French 17 industry codes. Standard errors are clustered at the industry level, and p -values are reported in parentheses. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

Table 9
Long-short portfolio returns

Accuracy model	All	Illiquidity		Idiosyncratic risk	
		Low	High	Low	High
Newsworthiness	0.964* (0.058)	0.181 (0.799)	1.429** (0.033)	0.190 (0.717)	1.102 (0.231)
Journalist characteristics	1.014* (0.083)	0.044 (0.954)	0.585 (0.440)	0.782 (0.145)	0.340 (0.712)
Article characteristics	0.932* (0.098)	−0.292 (0.694)	1.004 (0.149)	0.427 (0.407)	1.068 (0.231)
Journalist fixed effects	0.707 (0.200)	−0.147 (0.846)	1.206* (0.065)	−0.165 (0.738)	1.009 (0.239)
Newspaper fixed effects	0.703 (0.196)	−0.012 (0.987)	1.438** (0.024)	0.221 (0.663)	1.609* (0.073)
Journalist and newspaper fixed effects	0.618 (0.295)	−0.314 (0.711)	1.160* (0.080)	−0.174 (0.728)	1.619* (0.067)

This table reports monthly returns from an equally-weighted long-short portfolio, with a long position in targets of rumors that are predicted to be more accurate than the median accuracy rate and a short position in targets with a predicted accuracy less than the median accuracy rate. Accuracy is predicted using the logit models based on the characteristics listed under the column named Accuracy Model, after controlling for the Day 0 return. For each set of characteristics, the predictive model is based on column 2 of the respective table: Table 6 (target newsworthiness), Table 7 (journalist characteristics), Table 8 (article characteristics), Internet Appendix Table 10 (journalist fixed effects), Internet Appendix Table 13 (newspaper fixed effects), and unreported tests that combine journalist and newspaper fixed effects in one model. Firms are added to the portfolio on the day after the rumor is published, so the Day 0 return is not included in the long-short portfolio return. Firms are held for up to one year and the portfolio is rebalanced monthly. If less than five firms meet the criteria for the long or short portfolio, the long-short portfolio return is assumed to be zero. The monthly returns presented here are the daily returns compounded 20 days. Newey and West (1987) p -values are in parentheses. We follow Newey and West (1994) to compute lags. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

5.9 Table 9: Long-short portfolio returns

Read. Portfolios sorted by predicted accuracy earn positive monthly long-short returns. Using newsworthiness to predict accuracy generates about 0.964% per month. Journalist characteristics generate about 1.014% per month. Article characteristics generate about 0.932% per month.

The returns are stronger among hard-to-arbitrage stocks. For example, the newsworthiness-based strategy earns 1.429% per month among high-illiquidity stocks but only 0.181% among low-illiquidity stocks.

Economic message. The mispricing is economically meaningful and is concentrated where arbitrage is costly. This supports a behavioral-limits-to-arbitrage interpretation rather than a pure riskless trading opportunity.

5.10 Figures 2–4: Reversals, shorting, and institutional trading

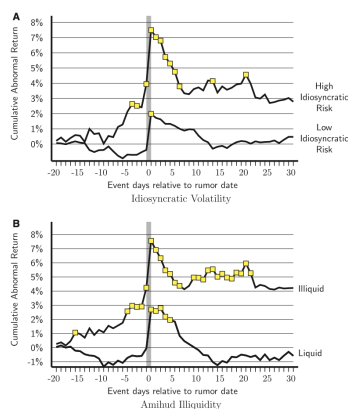


Figure 2
Reversals following inaccurate rumors by limits to arbitrage
This figure presents the time series relative to the date of the first publication of a merger rumor of cumulative abnormal returns of targets in rumors that do not come true within one year of the publication of the rumor. Cumulative abnormal returns are the cumulative sum of daily abnormal returns, calculated as the firm's daily return minus the value-weighted CRSP index. The squares indicate cumulative returns that are statistically significant at the 10% level. Panel A presents abnormal returns of targets with above- or below-median idiosyncratic volatility. Panel B presents abnormal returns for targets with above or below median Amihud illiquidity.

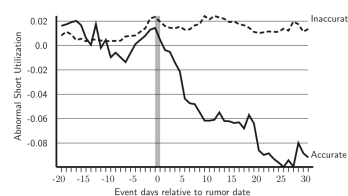


Figure 3
Abnormal shorting utilization by rumor accuracy in event time
This figure presents the time series relative to the date of the first publication of a merger rumor of average abnormal shorting utilization of targets in rumors. Shorting utilization is the fraction of shares on loan divided by shares available for short positions. Abnormal short utilization is daily short utilization minus the average daily utilization over the six month period that ends twenty days before the rumor is published. Data are from Market from 2006 to 2011.

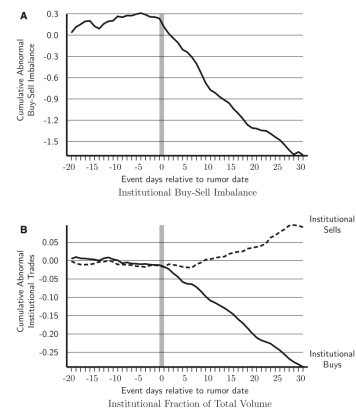


Figure 4
Daily institutional trading in event time
This figure presents the time series, relative to the date of the first publication of a merger rumor, of the cumulative abnormal institutional buy-sell imbalance of trading in targets of rumors in panel A and institutional buys and sells normalized by total volume in panel B. Buy-sell imbalance is the daily number of shares purchased minus the number of shares sold by all institutional traders in the ANcerno database, normalized by the sum of buys and sells. Abnormal buy-sell imbalance is the daily buy-sell imbalance minus the average daily imbalance over the six month period that ends 20 days before the rumor is published. Panel B presents the cumulative abnormal fraction of total trades by institutions in ANcerno. The fraction of total volume is all ANcerno trades/CRSP trading volume, where ANcerno trades are separated into buys and sells. Abnormal trading volume is normalized as above. Data are from the period 2000–2011.

Figure 2 read. After inaccurate rumors, high-idiosyncratic-volatility targets take about 21 days to fully reverse, while low-idiosyncratic-volatility targets take about one day. Illiquid targets take about 22 days to reverse, while liquid targets take about five days.

Figure 3 read. Short utilization falls after accurate rumors but is relatively flat after inaccurate rumors. Short sellers appear to reduce short exposure when a rumor is likely to be true.

Figure 4 read. Institutional investors are net sellers after rumor publication. Total market trading volume increases much more than ANcerno institutional volume, suggesting that non-institutional or retail trading drives much of the immediate reaction.

Economic message. Sophisticated investors are not the main buyers chasing rumor targets. The price impact is consistent with retail attention and limited processing of credibility signals.

5.11 Table 10: Run-up and total takeover premium

Read. Rumored deals have higher pre-announcement run-ups but lower announcement-period returns:

Dependent variable	Rumor coefficient	Message
Target CAR $[-42, -1]$	about +0.077	Higher pre-announcement run-up
Target CAR $[0, +5]$	about -0.078	Lower announcement markup
Target CAR $[-42, +5]$	about 0.000	No higher total premium

Economic message. Rumors move target prices before official announcements, but acquirers appear to discount this run-up when setting the final premium. Thus, rumors affect price dynamics and negotiation narratives, but not the average total takeover premium paid.

6 Short summary

Ahern and Sosyura study whether sensationalism in financial media affects accuracy and asset prices. They focus on merger rumors because the truth of the rumor can be evaluated ex post by observing whether the target receives a public takeover bid within one year.

The authors hand-collect 501 merger rumors from 2000–2011 and identify the first article reporting each rumor. Only 33.3% of rumors come true. Rumor targets are not random: they are much more likely than actual merger targets to be public firms, large firms, valuable brands, advertising-intensive firms, and consumer-facing firms.

The central finding is that attention-grabbing targets are less likely to be associated with accurate rumors. This is consistent with the idea that the business press may publish speculative rumors about high-readership firms even when the evidence is weaker.

Accuracy is also predicted by who writes the story and how the story is written. Older journalists, journalists trained in journalism, and journalists with target-industry expertise are more accurate. Articles with vague modal language are less accurate; articles with specific prices, named bidders, or concrete negotiation stages are more accurate.

Stock prices respond strongly to rumors. The average target earns about 4.3% on the rumor date. Accurate rumors earn about 6.9%, while inaccurate rumors still earn about 3.0%. Inaccurate rumors reverse afterward, generating an average post-rumor decline of about 2.7% over days $[+1, +10]$.

The key asset-pricing result is that investors do not fully process public credibility signals. News-worthiness, journalist traits, and article text continue to predict accuracy even after controlling for day-0 returns and estimated deal likelihood. Portfolios sorted on predicted accuracy earn positive long-short returns, especially among illiquid and high-idiosyncratic-risk stocks.

The paper also shows that merger rumors have real effects on merger negotiations. Merger agreements often discuss rumors and their price impact. In the data, rumored deals have higher run-ups but lower announcement markups, so total takeover premiums are not higher.

7 Conclusion

7.1 Core conclusion

Financial media coverage is shaped by the incentive to attract attention. In the context of merger rumors, this incentive is associated with lower accuracy for stories about large, prominent, high-

readership firms.

7.2 Economic intuition

Readers like stories about firms they recognize. Newspapers therefore have an incentive to cover sensational rumors about famous companies. But those rumors are less likely to be true because the readership payoff from publishing them is high even when the evidentiary basis is weak.

7.3 Main empirical contribution

The paper builds a novel hand-collected dataset that links rumor articles, target characteristics, journalist biographies, article text, newspaper traits, stock returns, short-selling activity, institutional trading, and merger outcomes. This allows the authors to study not just whether media coverage moves prices, but whether the coverage is accurate.

7.4 Main asset-pricing contribution

The paper shows that investors respond to rumors but do not fully process credibility signals. The result is an initial overreaction, predictable reversals, and return predictability based on public but non-salient information.

7.5 Main media-economics contribution

The paper challenges the view of the press as a passive information conduit. The business press actively selects stories, and that selection is partly shaped by consumer demand for attention-grabbing news.

7.6 Takeaway

A merger rumor is not just information about a possible deal. It is also a media product. To interpret it correctly, investors must ask not only *what* the article says, but *why* it was published, *who* wrote it, and *how* credible its wording and details are.